

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 7

End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modeling. The experiment progresses from a fully connected autoencoder to a convolutional autoencoder, a denoising convolutional autoencoder and finally a variational autoencoder (VAE). The learned latent representations are visualized and the models are evaluated using reconstruction metrics.

2. Learning Outcomes

After completing this experiment, the following were studied and implemented:

- encoder–latent–decoder structure of an autoencoder;
- image preparation for reconstruction learning;
- fully connected and convolutional autoencoders;
- controlled image corruption and denoising;
- MSE, MAE and SSIM for reconstruction evaluation;
- deterministic and probabilistic latent representations;
- the VAE reparameterization trick and KL-divergence loss;
- latent-space visualization, sampling and interpolation; and
- reconstruction-error analysis and high-error image inspection.

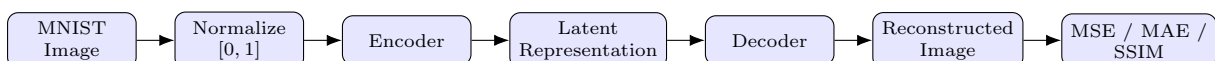
3. Dataset and Experimental Setup

Dataset: MNIST Handwritten Digit Dataset. MNIST contains grayscale handwritten digits from 0 to 9. Each image has spatial dimensions $28 \times 28 \times 1$. The laboratory execution used a fixed subset of 10,000 training images and 2,000 test images, as recommended by the experiment manual. Pixel values were normalized from $[0, 255]$ to $[0, 1]$. Labels were not supplied as reconstruction targets; the input image itself was used as the target.

Training images	10,000
Test images	2,000
Image size	$28 \times 28 \times 1$
Flattened size	784
Batch size	128
FC/CAE epochs	20
VAE epochs	30
Optimizer	Adam
Learning rate	10^{-3}

The TensorFlow runtime reported version 2.20.0 and detected one GPU. The same train/test data protocol was maintained across the reconstruction models.

4. Overall Experimental Pipeline



Four related studies were performed: fully connected AE, convolutional AE, denoising CAE and VAE. An additional latent-dimension study was performed for the fully connected autoencoder.

5. Autoencoder Theory

An autoencoder learns a reconstruction mapping

$$z = f_{\theta}(x), \quad \hat{x} = g_{\phi}(z).$$

The reconstruction objective is to minimize the difference between x and $\hat{x}$. For squared reconstruction error,

$$L_{rec} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2.$$

The latent vector is a compressed representation of the input.

6. Github Link:

<https://github.com/PranavVarshin/DeepLearning/tree/main/Lab-7>

7. Fully Connected Autoencoder

The images were flattened from 28×28 to 784 values. The implemented architecture was

$$784 \rightarrow 128 \rightarrow 32 \rightarrow d_z \rightarrow 32 \rightarrow 128 \rightarrow 784,$$

with ReLU activations in hidden layers and sigmoid activation at the output. The main model used $d_z = 16$.

The model was trained for 20 epochs using Adam with learning rate 10^{-3} , batch size 128 and binary cross-entropy loss. Ten percent of the training subset was used for validation.

7.1 Reconstruction and Convergence

Figure 1 shows original images, FC-AE reconstructions and absolute reconstruction errors.

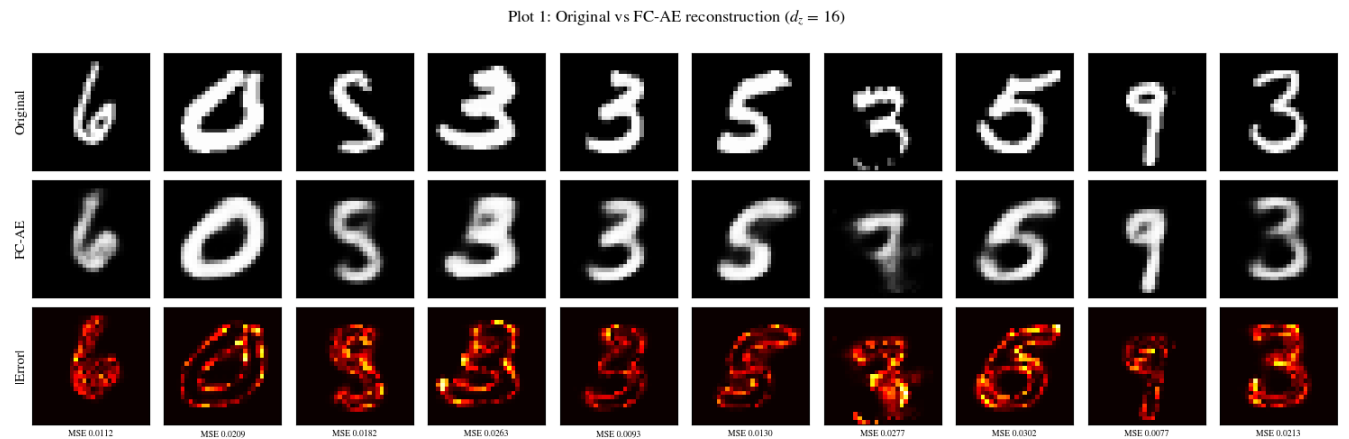


Figure 1: Original images, FC-AE reconstructions and absolute error maps.

The final training loss was 0.1217 and validation loss was 0.1226, giving a small gap of 0.0009. The curves decrease steadily over the 20 epochs, indicating convergence without a clear validation-loss divergence.

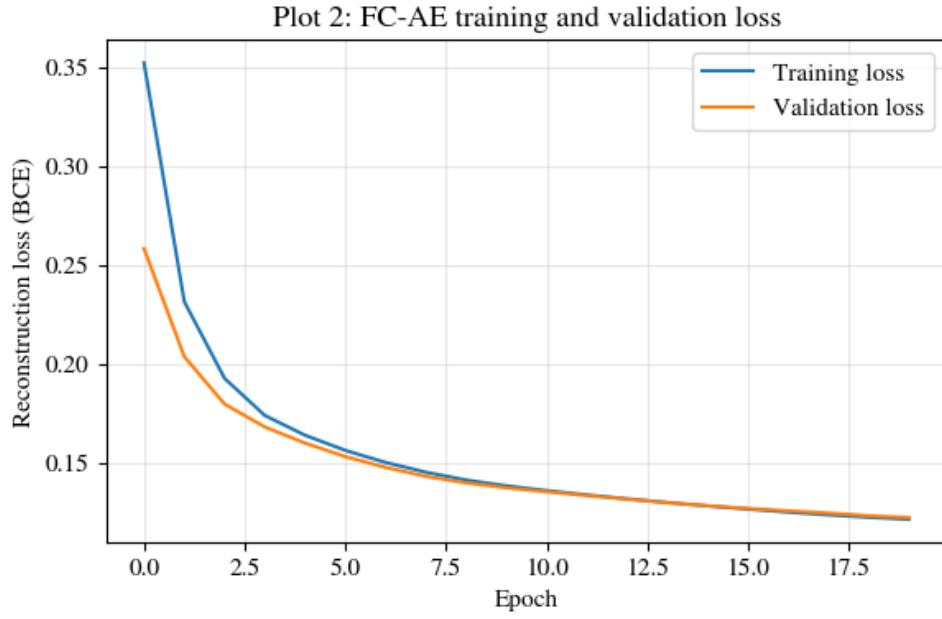


Figure 2: FC-AE training and validation reconstruction loss.

7.2 Test Metrics

Test MSE	0.01992
Test MAE	0.05442
Mean SSIM	0.7749
Parameters	211,040
Training time	12.0 s

7.3 Latent-Dimension Study

The FC-AE was repeated with $d_z \in \{2, 8, 16, 32\}$.

Latent dimension	MSE	SSIM	Parameters
2	0.04718	0.4729	210130
8	0.02595	0.7107	210520
16	0.02025	0.7712	211040
32	0.01919	0.7814	212080

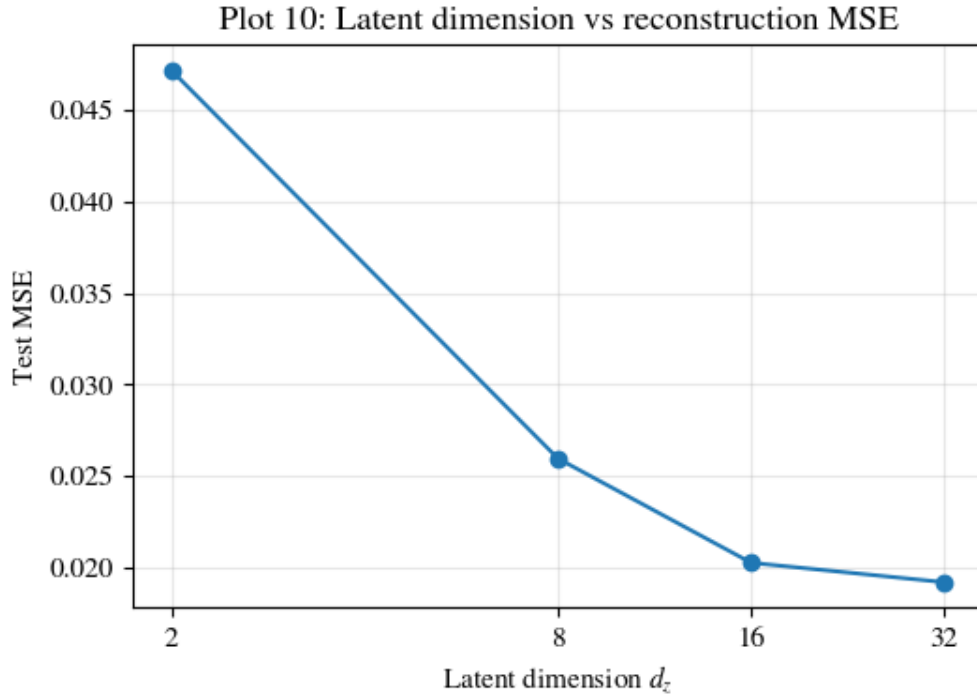


Figure 3: Latent dimension versus reconstruction MSE and representative reconstructions.

Increasing the latent dimension improved reconstruction quality. The largest change occurred from 2 to 8 dimensions, while the improvement from 16 to 32 was comparatively small. This demonstrates the compression–reconstruction trade-off.

8. Convolutional Autoencoder

The convolutional autoencoder preserved the spatial structure of the images. The implemented architecture was

$$\text{Conv32} \rightarrow \text{MaxPool} \rightarrow \text{Conv64} \rightarrow \text{MaxPool} \rightarrow \text{Conv64}$$

for the encoder, followed by upsampling, convolution and a sigmoid output for the decoder. The latent feature map contained $7 \times 7 \times 64 = 3136$ values.

The model used the same 20-epoch training protocol as the FC-AE.

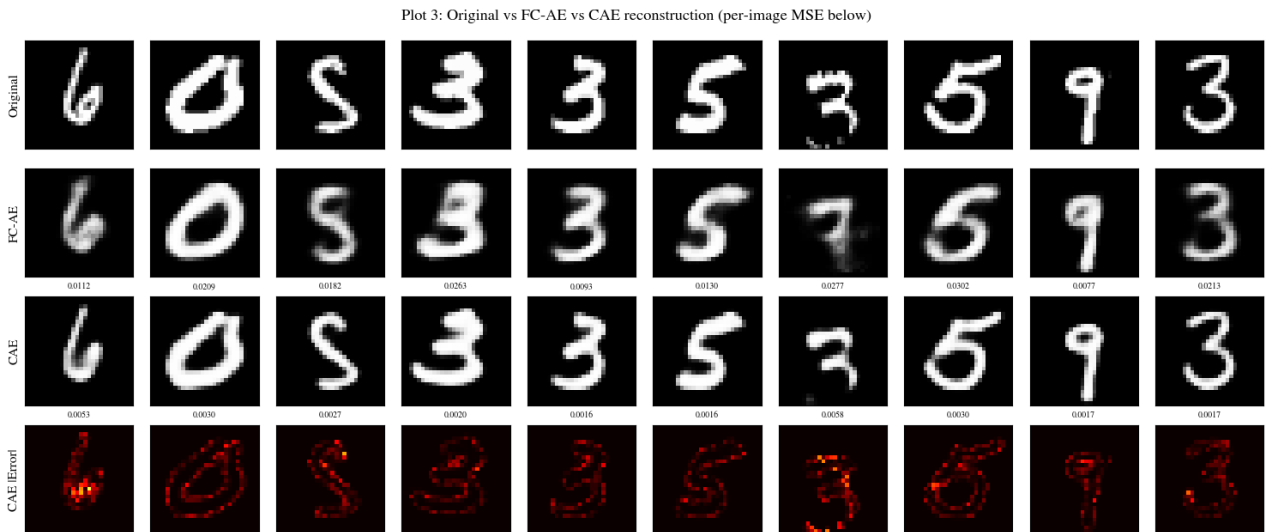


Figure 4: FC-AE versus convolutional AE reconstruction comparison.

Model	MSE	MAE	SSIM	Parameters
FC Autoencoder	0.01992	0.05442	0.7749	211040
Conv. Autoencoder	0.00264	0.01554	0.9725	74497

The convolutional autoencoder reduced MSE by approximately 86.7% relative to the FC-AE and increased SSIM by 0.1976. This is consistent with the benefit of preserving local spatial structure through convolution.

9. Denoising Convolutional Autoencoder

A denoising autoencoder receives a corrupted image $\tilde{x}$ and learns to reconstruct the clean image x :

$$\tilde{x} \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}.$$

Two corruption types were tested.

9.1 Gaussian Noise

Gaussian corruption was generated using

$$\tilde{x} = x + n, \quad n \sim \mathcal{N}(0, \sigma^2),$$

with $\sigma \in \{0.1, 0.2, 0.3\}$. Values were clipped to $[0, 1]$.

9.2 Salt-and-Pepper Noise

For salt-and-pepper corruption, a proportion $p \in \{0.05, 0.10, 0.20\}$ of pixels was replaced with values close to 0 or 1. The denoising CAEs used the same convolutional architecture and training protocol as the plain CAE. The headline denoising model used Gaussian noise with $\sigma = 0.2$.

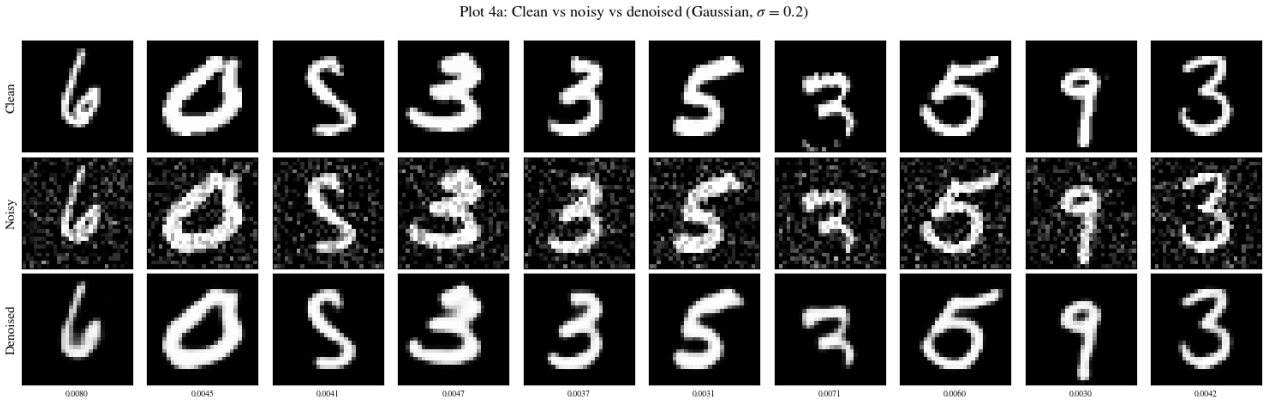


Figure 5: Clean, noisy and denoised MNIST images for Gaussian noise.

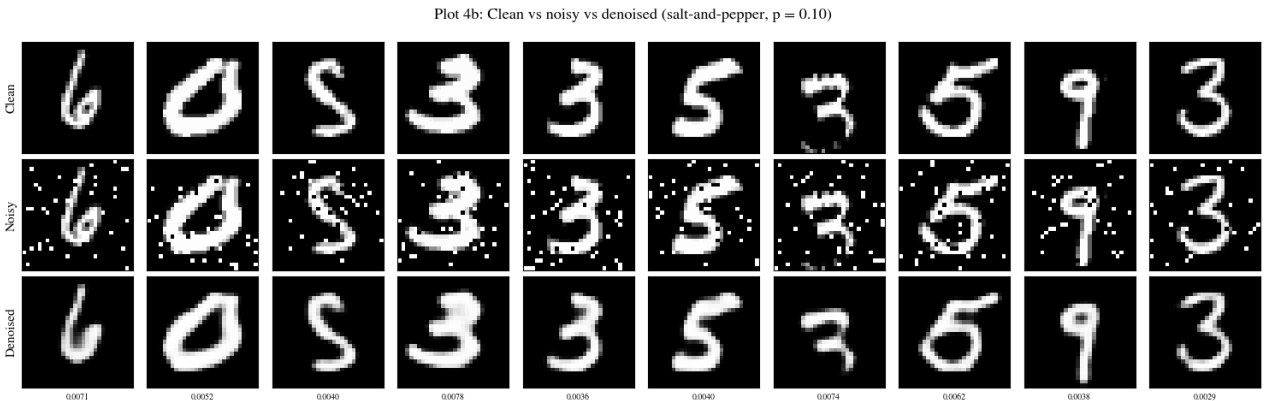


Figure 6: Clean, noisy and denoised MNIST images for salt-and-pepper noise.

9.3 Denoising Results

Level	Gaussian noise			Denoised output
	MSE	MAE	SSIM	
0.1	0.00315	0.01734	0.9649	
0.2	0.00444	0.02111	0.9480	
0.3	0.00659	0.02641	0.9222	

Salt-and-pepper p	MSE	MAE	SSIM
0.05	0.00379	0.01871	0.9587
0.10	0.00516	0.02210	0.9441
0.20	0.00782	0.02850	0.9121

Plot 5: Noise level vs MSE, MAE and SSIM (Gaussian)

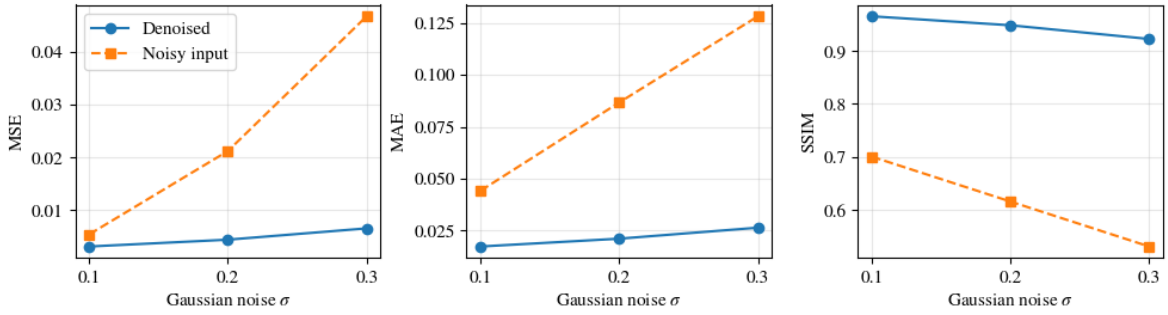


Figure 7: Gaussian noise level versus denoising reconstruction quality.

Plot 5: Noise level vs MSE, MAE and SSIM (salt-and-pepper)

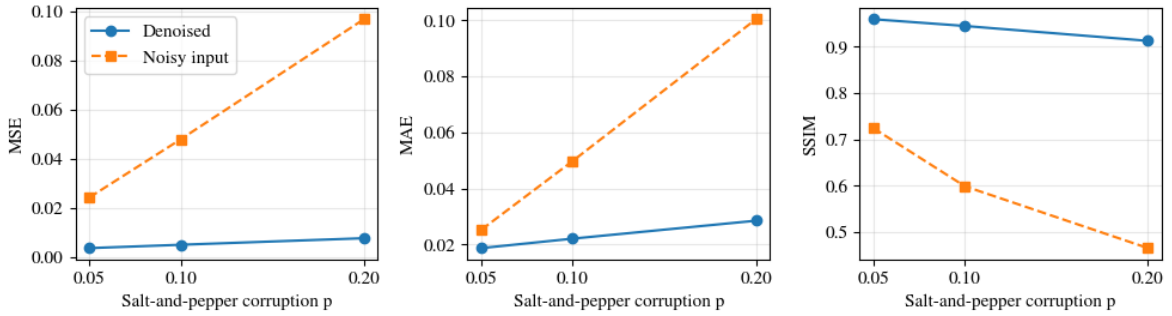


Figure 8: Salt-and-pepper noise level versus denoising reconstruction quality.

As corruption increased, denoising MSE and MAE increased while SSIM decreased. However, the denoised outputs remained substantially better than the corresponding noisy inputs, showing that the model recovered the major digit structure.

10. Variational Autoencoder

Unlike a conventional autoencoder, the VAE learns a probability distribution in latent space:

$$q_{\phi}(z|x) = \mathcal{N}(\mu(x), \text{diag}(\sigma^2(x))).$$

The encoder produced μ and $\log \sigma^2$ for a two-dimensional latent representation. Sampling used the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

The VAE used a convolutional encoder with two strided convolution layers, a dense layer and two latent heads. The decoder mapped a two-dimensional latent vector back to a $28 \times 28 \times 1$ image using a dense layer and transposed convolutions.

The loss consisted of reconstruction and KL-divergence terms:

$$L_{VAE} = L_{rec} + L_{KL},$$

where

$$D_{KL}(q_\phi(z|x)||p(z)) = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2).$$

The prior was $p(z) = \mathcal{N}(0, I)$.

The VAE was trained for 30 epochs with Adam, learning rate 10^{-3} and batch size 128. It contained 134,165 trainable parameters and required 26.9 seconds.

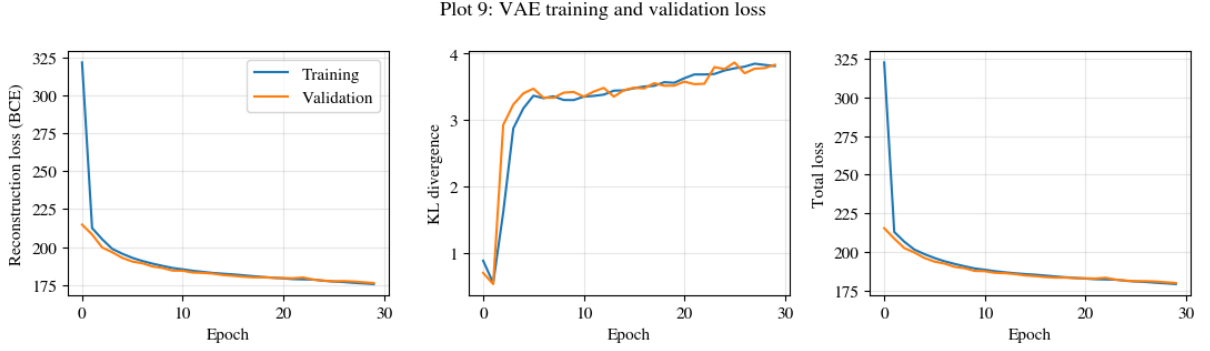


Figure 9: VAE reconstruction, KL and total training/validation losses.

The final training reconstruction, KL and total losses were 175.55, 3.81 and 179.37 respectively. The corresponding validation values were 176.25, 3.83 and 180.08.

10.1 VAE Test Metrics

Reconstruction loss	176.266
KL loss	3.857
Total loss	180.123
Test MSE	0.05474
Test MAE	0.12666
Mean SSIM	0.3661
Parameters	134,165
Training time	26.9 s

11. VAE Latent Space and Generation

The two-dimensional latent representation was visualized using the encoded means (z_1, z_2) . The VAE was not trained using labels; labels were used only to colour the visualization and calculate the auxiliary 5-NN statistic. The mean of the encoded means was approximately $[-1.01, -0.016]$ and the standard deviations were approximately $[0.757, 0.038]$. The mean encoded standard deviations were $[0.038, 0.846]$. The KL loss per latent dimension was approximately $[3.813, 0.044]$, with total KL loss 3.857.

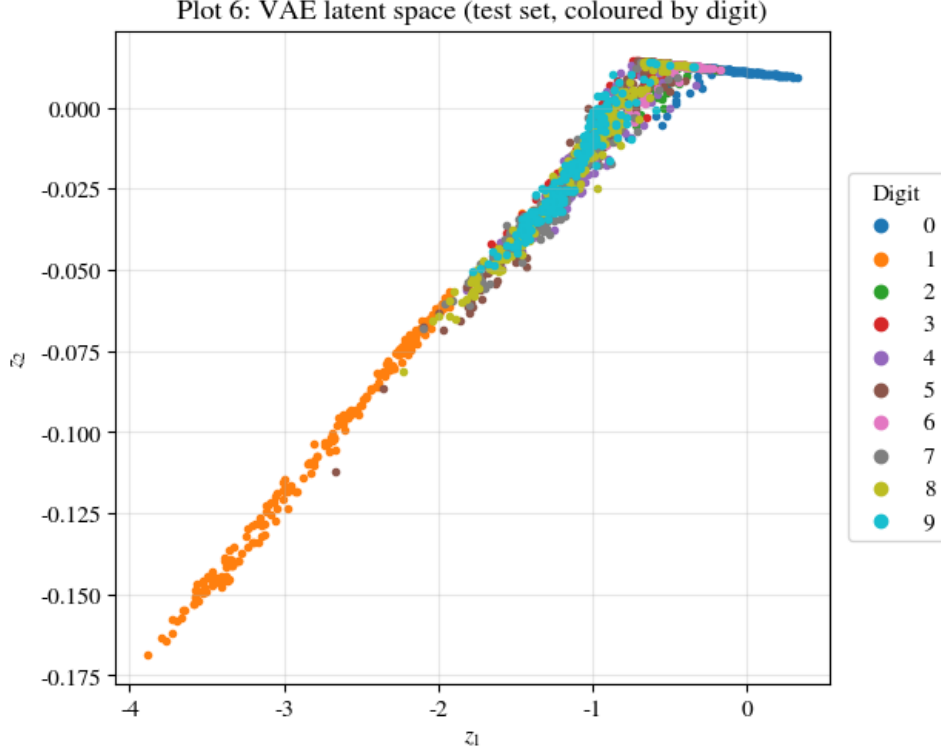


Figure 10: Two-dimensional VAE latent-space visualization.

The auxiliary 5-NN digit accuracy in the 2-D latent space was 0.412, indicating substantial overlap among digit classes in the learned representation.

11.1 Random Generation

New latent vectors were sampled from $\mathcal{N}(0, I)$ and passed through the decoder without an input image. At least 25 samples were generated.

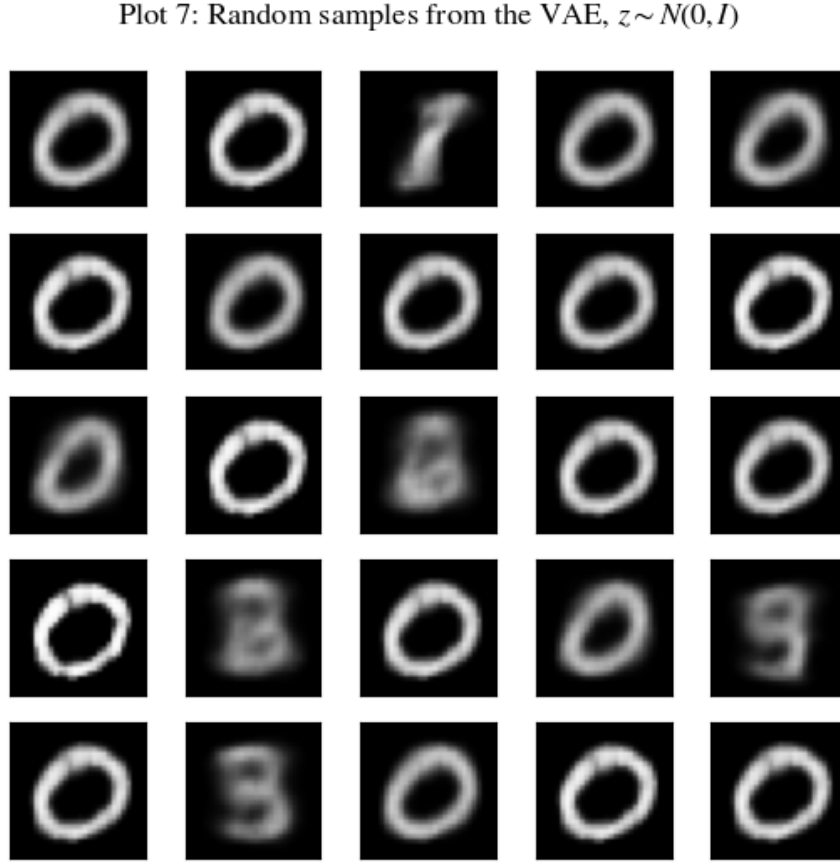


Figure 11: Randomly generated MNIST images from the VAE prior.

The generated samples demonstrate that the VAE can produce digit-like images from latent samples. Some samples are clearer than others, reflecting the probabilistic nature and limited two-dimensional latent capacity.

11.2 Latent-Space Interpolation

For two latent vectors z_A and z_B , interpolation was performed using

$$z(\alpha) = (1 - \alpha)z_A + \alpha z_B, \quad \alpha \in [0, 1].$$

Values of α from 0 to 1 in steps of 0.1 were decoded.

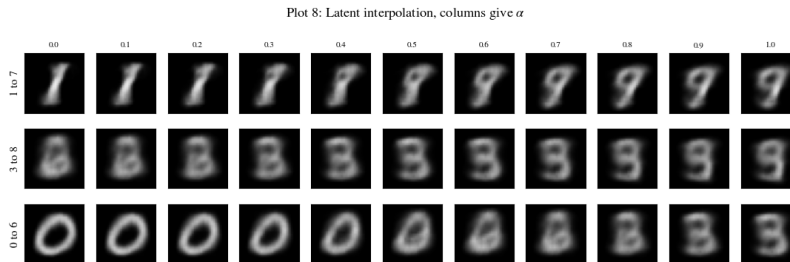


Figure 12: Latent-space interpolation examples.

The mean consecutive-frame MSE values were 0.0009 for the 1-to-7 pair, 0.0005 for 3-to-8 and 0.0020 for 0-to-6. The decoded images changed progressively between endpoints, demonstrating the continuous structure of the learned latent space.

12. Deterministic versus Probabilistic Latent Representations

A separate FC-AE with $d_z = 2$ was trained to compare a deterministic latent representation with the VAE latent space.

The FC-AE latent standard deviations were approximately $[13.72, 10.43]$ with an auxiliary 5-NN accuracy of 0.588. The VAE latent standard deviations were approximately $[0.76, 0.04]$ with 5-NN accuracy of 0.412.

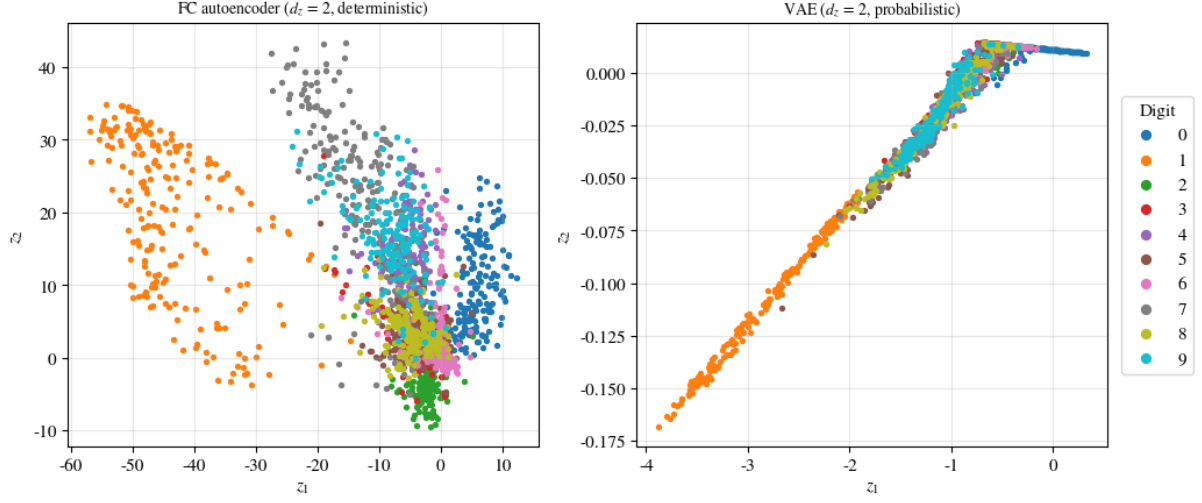


Figure 13: Comparison of deterministic FC-AE and probabilistic VAE latent spaces.

The FC-AE learns a direct deterministic code for each image, whereas the VAE constrains the latent distribution toward a structured prior and supports sampling of new points.

13. Reconstruction Error Distribution

For each test image, the reconstruction error was calculated as

$$e_i = \frac{1}{D} \sum_{j=1}^D (x_{ij} - \hat{x}_{ij})^2.$$

The resulting distributions were analyzed for the FC-AE, CAE and VAE.

Model	Mean	Median	SD	Min	P90	P99	Max	$> \mu + 2\sigma$
FC-AE	0.01992	0.01894	0.01038	0.00184	0.03375	0.04822	0.07570	73
CAE	0.00264	0.00243	0.00137	0.00039	0.00437	0.00724	0.01145	85
VAE	0.05474	0.05496	0.01890	0.00513	0.07681	0.10548	0.14258	46

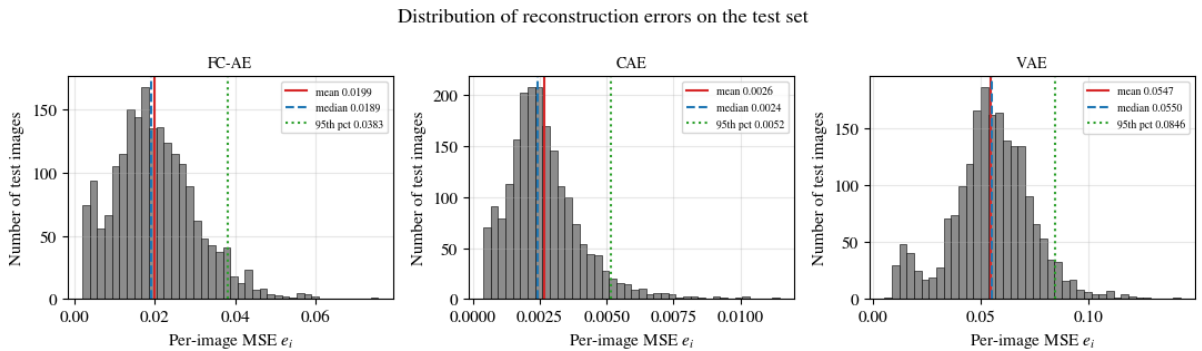


Figure 14: Distribution of per-image reconstruction errors.

The CAE has the lowest typical reconstruction error and a relatively concentrated error distribution. The VAE has a higher error distribution, consistent with the strong compression imposed by its two-dimensional probabilistic latent space.

14. High-Error Image Analysis

The five highest-error samples were identified for each model. For the FC-AE, the largest-error samples included digit 5 (error 0.0757), digit 8 (0.0603 and 0.0592), digit 6 (0.0588) and digit 4 (0.0586). For the CAE, the largest errors ranged from 0.0093 to 0.0114. For the VAE, the five largest errors ranged from 0.1201 to 0.1426.

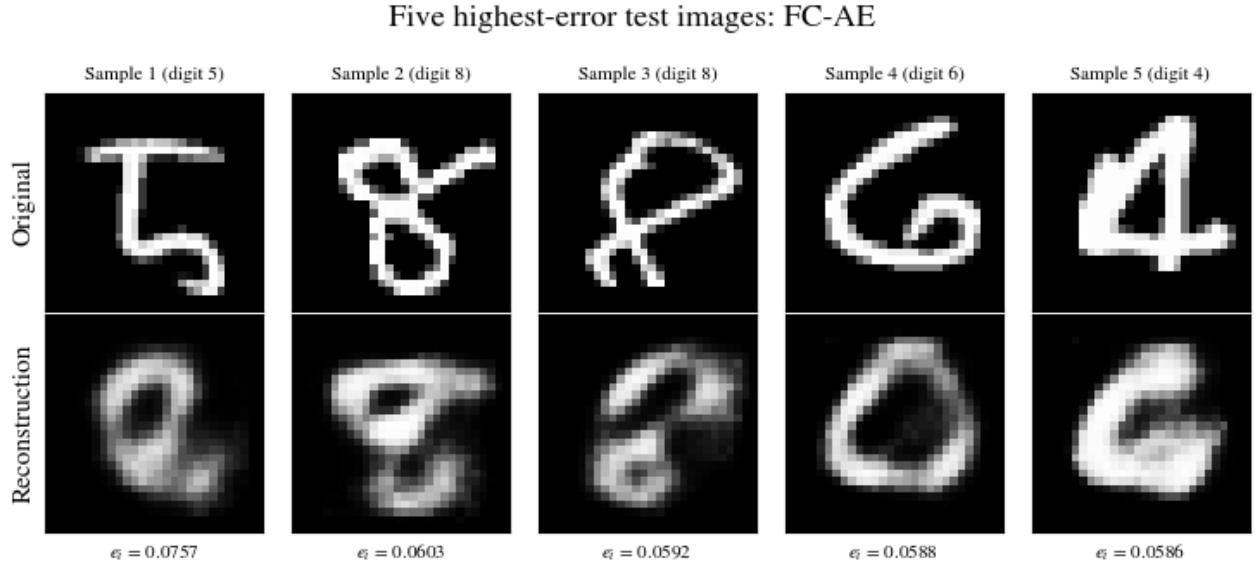


Figure 15: Highest-error examples for the FC-AE.

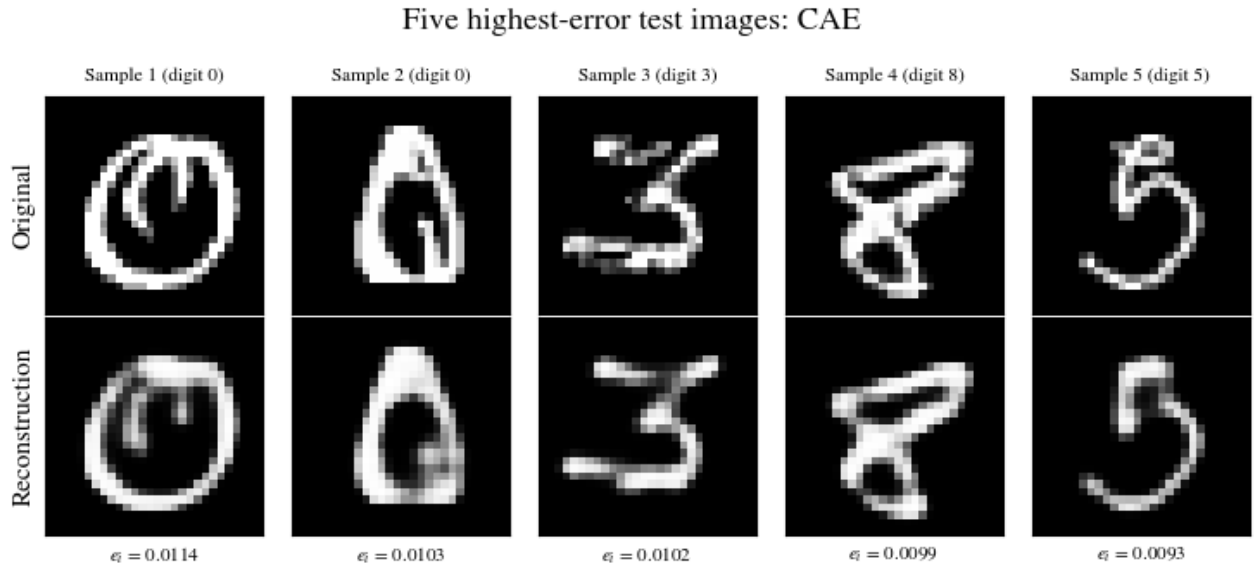


Figure 16: Highest-error examples for the convolutional autoencoder.

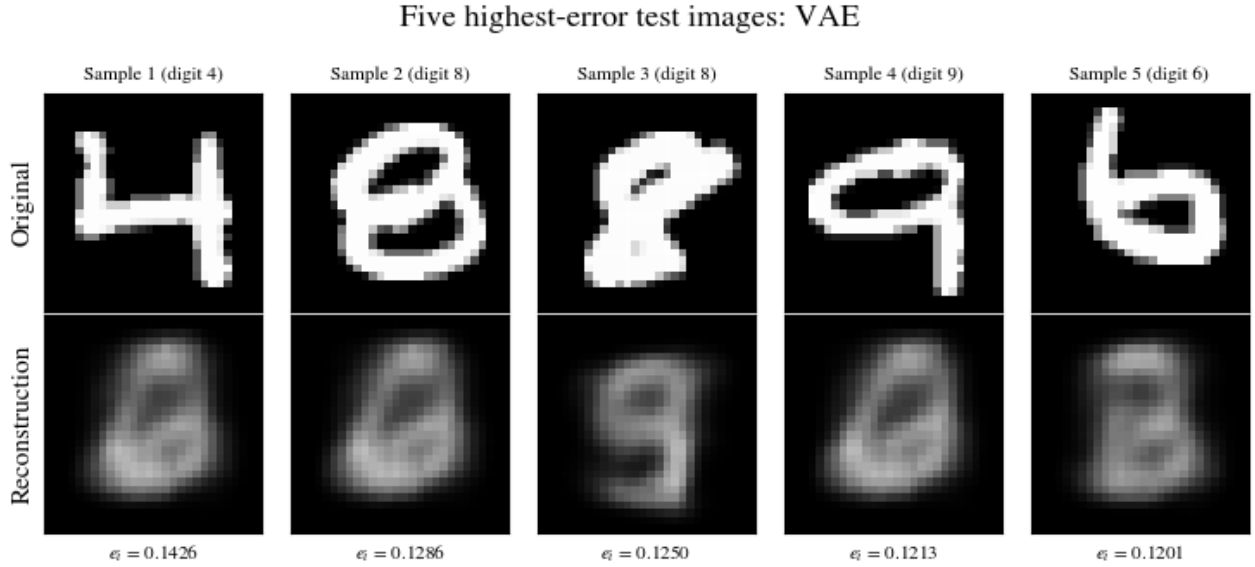


Figure 17: Highest-error examples for the VAE.

The digit-wise analysis showed that digit 8 was the hardest class for the FC-AE and CAE, while digit 2 was hardest for the VAE. Digit 1 had the lowest mean reconstruction error for all three models. The Spearman correlation between error and total image intensity was 0.39 for FC-AE, 0.31 for CAE and 0.63 for VAE.

15. Consolidated Results

Model	MSE	MAE	SSIM	Parameters	Time (s)
FC Autoencoder	0.01992	0.05442	0.7749	211040	12.0
Conv. Autoencoder	0.00264	0.01554	0.9725	74497	20.7
Denoising CAE	0.00444	0.02111	0.9480	74497	17.8
VAE	0.05474	0.12666	0.3661	134165	26.9

The plain CAE achieved the lowest test MSE and highest mean SSIM among the reconstruction models in this execution. The denoising CAE also retained high structural similarity while handling corrupted inputs. The FC-AE was faster to train but used more parameters and produced larger reconstruction errors. The VAE had a different objective because its total loss includes KL divergence in addition to reconstruction loss.

16. Required Inferences

- **Effect of latent dimension:** Increasing d_z from 2 to 32 reduced MSE from 0.04718 to 0.01919 and increased SSIM from 0.4729 to 0.7814.
- **FC-AE vs CAE:** The CAE preserved spatial locality and produced substantially lower reconstruction error with fewer parameters in this experiment.
- **Gaussian noise:** Increasing σ from 0.1 to 0.3 increased denoised MSE and MAE and reduced SSIM.
- **Denoising ability:** The denoising CAE recovered major digit structure and improved all three reported metrics relative to the noisy inputs.
- **VAE latent distribution:** The VAE organizes information in a probabilistic latent space constrained by KL divergence toward a standard normal prior.
- **Latent-space structure:** The 2-D VAE representation contains overlapping digit regions, as reflected by the auxiliary 5-NN accuracy of 0.412.
- **Generation:** Sampling from $\mathcal{N}(0, I)$ produced digit-like images with visible variation, although some generated samples were ambiguous.
- **Interpolation:** Decoding linearly interpolated latent vectors produced gradual changes between selected end-points.
- **Reconstruction error and visual quality:** Higher per-image MSE generally corresponds to more visible reconstruction differences, motivating inspection of high-error samples.
- **Deterministic vs probabilistic representation:** The FC-AE maps each input to a deterministic code, while the VAE learns distribution parameters and enables sampling and generation.

17. Conclusion

The experiment demonstrated a complete progression from deterministic image reconstruction to denoising and probabilistic generative modeling. The fully connected autoencoder successfully learned compact representations, while the convolutional autoencoder substantially improved reconstruction quality by preserving spatial structure. The denoising CAE recovered meaningful digit structure from both Gaussian and salt-and-pepper corruption. The VAE learned a two-dimensional probabilistic latent space, generated new digit-like images and supported smooth latent interpolation. Overall, the results illustrate the different objectives and behavior of deterministic reconstruction models, denoising models and variational generative models.

18. References

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6. Keras Documentation: <https://keras.io>
7. scikit-image Documentation: <https://scikit-image.org>

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